

Capstone Exercise & Course Wrap-Up

Latihan Kemuncak & Penutup Kursus

Every skill from Modules 1-8, run once more as one team, from business requirements to production monitoring, on the ShopFast quality trail built across both days.

2-Day Hands-On SQA Workshop

Aligned to KRISA 2.0 (JDN)

TIME

16:30 - 17:30

DURATION

1h 00m

FORMAT

Team enterprise case study

PHASE COVERAGE

All 5 phases

LAB TYPE

Team steps (10)

09

Jadual slot: Modul 9: Latihan Kemuncak dan Penutup

Run sheet / session timing

Hari 2, 16:30 hingga 17:30 · 60 minit · kira-kira 80% amali

10 min

Taklimat kes

30 min

Kemuncak berpasukan, AI sebagai rakan

10 min

Demo AI W09 dan per...

8 min

Pembentangan

2 min

Penutup

Masa	Segmen	Tempoh
16:30 hingga 16:40	Taklimat kes	10 minit
16:40 hingga 17:10	Kemuncak berpasukan, AI sebagai rakan	30 minit
17:10 hingga 17:20	Demo AI W09 dan persediaan	10 minit
17:20 hingga 17:28	Pembentangan	8 minit
17:28 hingga 17:30	Penutup	2 minit

KRISAv2: 6 fasa, 18 templat serahan, satu aliran DevOps

KRISAv2: 6 phases, 18 deliverable templates, one DevOps flow

Setiap fasa melalui Gate Review dengan senarai semak serahan dan Definition of Done (Bab 8, 8.3.1)

1Permulaan <i>Initiation</i> PPP / PPrISA02 D02 BRS RTM awal (T2.8) KRISAv2 Bab 2M1, M2	2Analisis <i>Analysis</i> D03 SRS D05 Pelan Migrasi D07 Pelan Integrasi KRISAv2 Bab 3M2	3Reka Bentuk <i>Design</i> D04 SDS D06 Spes. Migrasi D08 Spes. Integrasi KRISAv2 Bab 4M3	4Pembangunan <i>Development</i> D09 Pangkalan Data D10 Kod Sumber D11 Laporan Ujian Sistem KRISAv2 Bab 5M4, M5, M6	5Pengujian Penerimaan <i>Acceptance Testing</i> D12 Pelan Induk Pengujian D13 Pelan UAT/PAT D14 Laporan UAT/PAT KRISAv2 Bab 6M4, M6, M7	6Pelaksanaan <i>Implementation</i> D15 Laporan Migrasi D16 Laporan Penamatan Ujian D17 Manual, D18 Serahan KRISAv2 Bab 7M7, M8
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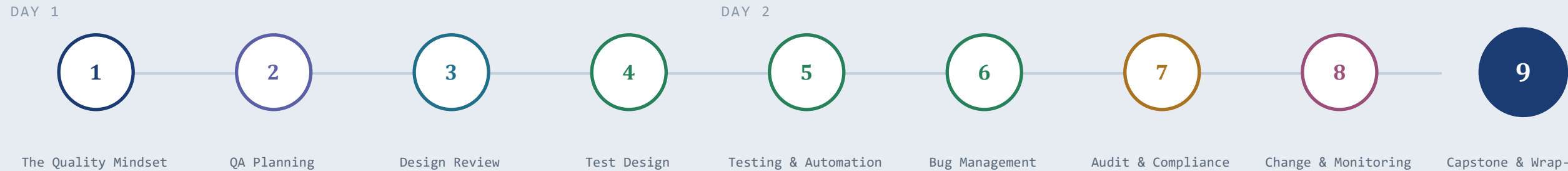
Bab 8 Pengelaran DevOps	Perancangan	Pra-Release dan Release: Analisis + Reka Bentuk + Pembangunan + Pengujian dalam sprint, CI/CD, Laporan Release	Pengoperasian: APM, log pemantauan
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PPrISA 2.0	Tadbir urus projek: DPP PPrISA01, PPP PPrISA02, Kualiti PPrISA05, Risiko PPrISA07, Pindaan PPrISA10/11, Status PPrISA16/17, Penamatan PPrISA18
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— WHERE THIS FITS

The workshop journey, nine stops, one project

Modules 1-8 practiced each SQA activity on the ShopFast checkout. Module 9 is the convergence point.



Modules 1-8 each ran one SQA activity, plan, review, design, test, defect-manage, audit, and monitor, on a single shared project. **Module 9** is where every team replays that full lifecycle end-to-end, independently, as proof of mastery.

SESSION SNAPSHOT

What this module is, at a glance

The final block of Day 2, a single guided team exercise that closes the workshop.

SCHEDULE	DURATION	PHASE PILL	DELIVERABLE
16:30-17:30 Day 2, closing block	1h 00m 60 minutes, team-run	All Phases Plan to Build-in to Verify to Assure to Sustain	Team presentation + feedback + certificate

"Bring every skill together: each team drives one enterprise case study from requirements to monitoring, then presents it."

THE CHALLENGE

One project, carried end to end

Every module runs on one shared project, the ShopFast checkout, so teams leave with one connected quality trail. The capstone is where that trail is walked in full, against an enterprise case study set by the trainer.

What your team already has

- QA Plan, Requirement Review report & RTM v0, Module 2
- Design Review checklist & Risk Register, Module 3
- Test Plan + 15-20 mapped test cases, Module 4
- API & UI automation scripts, Module 5
- Defect reports, triage board & metrics, Module 6
- Audit report, CAPA & Test Summary Report, Module 7
- Change Request, CI Quality Gate & dashboard, Module 8

What the capstone asks

Re-run the full nine-activity SQA lifecycle as one coherent narrative, requirements to planning to review to testing to defects to audit to change to monitoring, then present the trail as evidence of a release-ready quality process.

LEARNING OBJECTIVE

Bring every skill together.

Each team drives one enterprise case study from requirements to monitoring, then presents it to the full cohort.

Requirements to Planning

Review to Testing

Defects to Audit

Change to Monitoring

TEAM STEPS (10)

Requirements through test design

Steps 1-5 rebuild the foundation: understand the case, plan for quality, and design what you'll verify.

1

Review the case study business requirements.

Ref: Module 1-2

2

Develop a QA Plan & Risk Register.

Ref: Module 2 • 3

3

Conduct Requirement Review & Design Review.

Ref: Module 2 • 3

4

Build the RTM & Risk-Based Test Strategy.

Ref: Module 2 • 3

5

Design & execute functional + non-functional test cases.

Ref: Module 4 • 5

TEAM STEPS (10)

Execution through continuous quality

Steps 6-10 close the loop: run it, prove it, govern the next change, and keep watching it in production.

6

Log & prioritize defects (severity / priority).

[Ref: Module 6](#)

7

Produce a Test Execution Report & QA documentation.

[Ref: Module 6 • 7](#)

8

Perform a quality audit against ISO/IEC & organizational standards.

[Ref: Module 7](#)

9

Evaluate a production change request & rollback plan.

[Ref: Module 8](#)

10

Recommend monitoring metrics & continuous improvement actions.

[Ref: Module 8](#)

SAMPLE MODULE REFERENCES

Every capstone step traces back to a lab you already ran

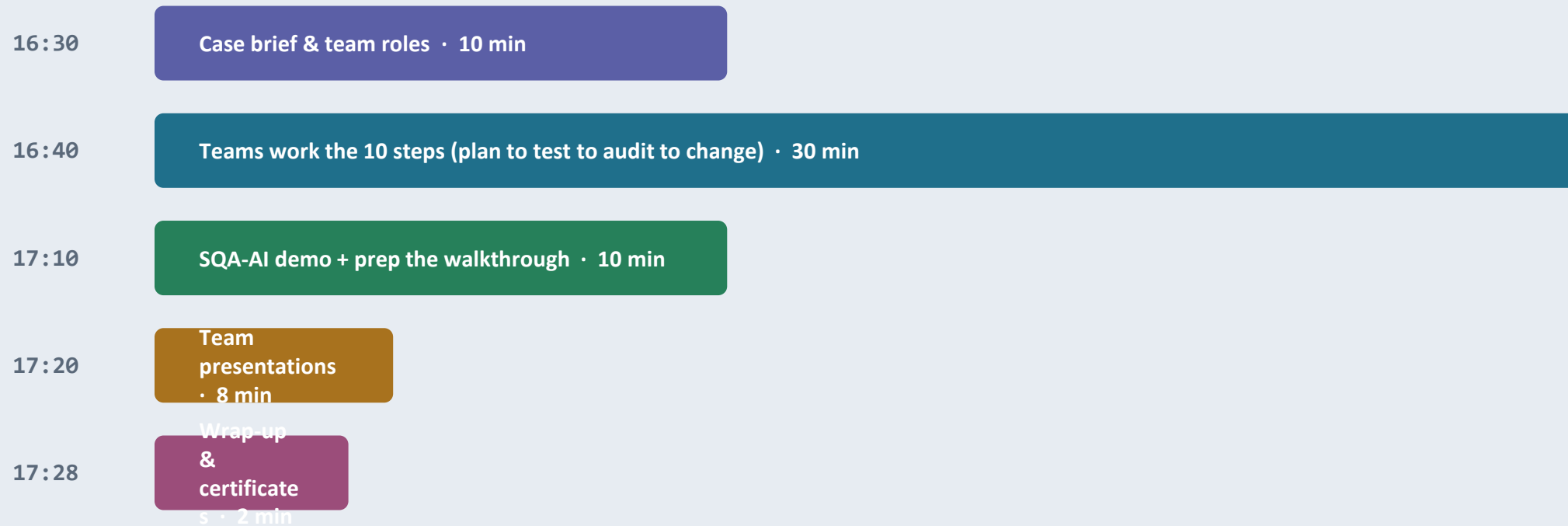
Use this map as a quick refresher, if a team gets stuck on a step, point them back to the matching module’s artifact.

Step	Team Action	Module	Origin Skill
1	Review case business requirements	1-2	The Quality Mindset / QA Planning
2	Develop QA Plan & Risk Register	2 · 3	QA Planning / Design Review
3	Requirement & Design Review	2 · 3	QA Planning / Design Review
4	Build RTM & Risk-Based Test Strategy	2 · 3	QA Planning / Design Review
5	Design & execute test cases	4 · 5	Test Design / Software Testing
6	Log & prioritize defects	6	Bug Management & Quality Metrics
7	Test Execution Report & QA docs	6 · 7	Bug Management / Audit & Compliance
8	Quality audit vs ISO/IEC & org standards	7	Audit, Compliance & Documentation
9	Evaluate change request & rollback plan	8	Change Management & Monitoring
10	Recommend monitoring metrics & CI actions	8	Change Management & Monitoring

— SUGGESTED FACILITATION FLOW

Pacing the 60-minute capstone

A trainer's suggested rhythm for the hour, adapt the split to team size and case complexity.



Suggested pacing for facilitation only, adapt to your cohort; not a fixed rule of the workshop.

What closes out the workshop

Team presentation + instructor feedback + participant reflection + course evaluation + certificate.



**Team
presentation**



**Instructor
feedback**



**Participant
reflection**



**Course
evaluation**

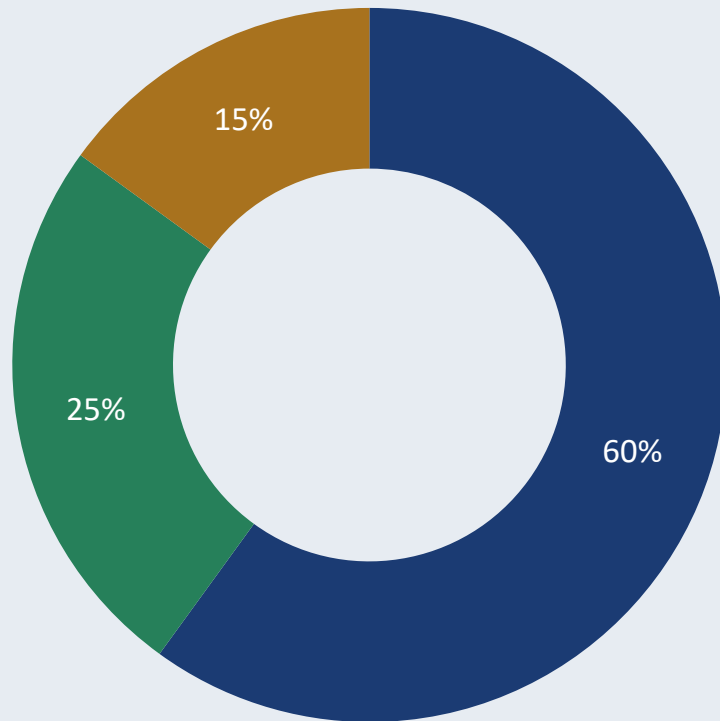


Certificate

HOW THE CAPSTONE IS ASSESSED

Assessment is the work itself

No written exam, participants are evaluated on the real artifacts produced across the two days, including the capstone walkthrough.

**60%, Lab artifacts**

12+ deliverables: QA Plan, RTM, test suite, automation, defect reports, audit & summary reports.

25%, Capstone walkthrough

Present the full ShopFast quality trail: from plan to a green CI gate & monitoring dashboard.

15%, Participation

Contribution to reviews, triage & the risk workshop across both days.

COURSE-WIDE OUTCOMES

What the capstone proves participants can now do

Aligned with the nine core SQA activities practiced across the workshop.

1 Establish & govern an enterprise SQA framework

2 Embed quality across the SDLC through Shift-Left practices

3 Design & run risk-based testing strategies

4 Apply test automation with confidence

5 Manage defects using meaningful quality metrics

6 Ensure compliance with recognized standards

7 Integrate quality into Agile & DevOps pipelines

8 Sustain reliability through change management, monitoring & continuous improvement

Capstone: The SQA Big Picture

Connecting the nine modules into one way of thinking about software quality

01 SQA on one slide

02 How the nine modules connect

03 A scientific, not religious, approach to methods

04 Maturity models as tools, not goals

05 Ten terms every practitioner must explain

06 What to study next

07 Principles to take back to work

SQA on one slide

Four elements, three control methods, and the loops that keep quality under control over time

SQA framework

Organisation procedures and standards: process standards and product standards.

SQP: planning

Select and tailor those standards for one project in a software quality plan.

SQC: reviews

In-process checks of products and processes, from desk checks to inspections.

SQC: tests

End-result verification of the product, from unit tests to run-for-record FAT and SAT.

SQC: audits

Periodic checks that the process standards are actually being followed.

SQM: metrics

Control and predictor metrics that steer the process and forecast product quality.

SCM

Baselines, change control and status accounting keep every change visible.

Improvement

Defect prevention, root cause analysis and Six Sigma close the loop.

How the nine modules connect

Module	Core ideas	Question it answers	Feeds into
M01 Quality Mindset	Quality, SQA vs SQC, fault, error and failure, V and V	What is quality, and who owns it?	All modules
M02 QA Planning	Standards, ISO 9001, IEEE 730, quality plans, review gates	Which standards apply, and how are they tailored?	M03, M07
M03 Design Review and Risk	Reviews, defect amplification, inspections, static analysis	How do we remove defects before test?	M04, M06
M04 Test Design	Test cases, oracles, harnesses, black box and white box limits	How do we design tests that find defects?	M05, M06
M05 Testing and Automation	Good automated tests, fixtures, TDD, CI, formal test runs	How do we run tests repeatably?	M06, M08
M06 Bug Management and Metrics	Reliability, a bug's life, metrics, defect prevention	What do defects and numbers tell us?	M07, M08
M07 Audit and Documentation	Audits, records, CMM and CMMI, ISO 9001 audit targets	Can we prove the process was followed?	M08
M08 Change and Continuous Quality	SCM, software evolution, Six Sigma, early warnings	How do we keep quality as software changes?	M02 in the next cycle
M09 Capstone	Big picture, evidence-based judgement, further study	How do the pieces fit together?	Your daily practice

Scientific, not religious, software engineering

Leveson: judge methods by evidence, not by how convincing they sound

Religious approach

- Accept a method on faith because it sounds right
- Follow gurus and join sects, one method or language against another
- Proof by vigorous handwaving
- Unsupported hypotheses and false analogies
- Adopt the latest trend because everyone else has

Scientific approach

- Ask for evidence: what data shows this method works, and where?
- Compare alternatives under comparable conditions
- Watch for the Hawthorne effect: attention alone can improve results for a while
- Learn history: what was tried before, what worked, and why
- Measure your own results before and after a change

Apply the same test to every new tool, AI assistant or maturity model: what is the evidence, and does it fit this project?

Maturity models are tools, not goals

“ *As a list of issues worth addressing in the course of software process improvement, the CMM is useful and benign.*

James Bach, postscript to *The Immaturity of CMM* (1999)

Bach's warning: problems begin when the model is adopted as a philosophy of good engineering.

A step-by-step level plan can give management an illusion of control and replace real learning.

Use a model's practices as a checklist of issues, then judge fit using your own evidence.

ShopFast goal: fewer escaped checkout defects and safer changes, not a certificate for its own sake.

Quiz: ten terms every practitioner must explain

Explain each term in one or two sentences, with a ShopFast example

- ☐ Software quality: the three layers of explicit requirements, standards and implicit expectations
- ☐ SQA, SQP, SQC and SQM: what each element does
- ☐ Fault, error and failure: explain using the quantity 1 to 10 rule
- ☐ Verification vs validation
- ☐ Formal technical review vs inspection
- ☐ Test oracle and test harness
- ☐ Software reliability and operational profile
- ☐ Control metric vs predictor metric
- ☐ Baseline and change control board (CCB)
- ☐ Nonconformance and corrective action

Pairs discuss for five minutes, then each pair explains one term to the whole room.

What to study next

Standards and bodies of knowledge, from general to specialised. Each can be mapped to the relevant KRISAv2 phase

Standard or body of knowledge	What it covers	Use it for
ISO/IEC 25010	Product quality model: quality characteristics and sub-characteristics	Writing measurable non-functional requirements
ISO/IEC/IEEE 29119 series	Software testing: concepts, processes, documentation, techniques	Test processes and documents (29119-3 supersedes IEEE 829)
IEEE 730	Software quality assurance processes	Planning and running SQA on a project
ISO 9001:2015	Quality management system requirements in 10 clauses	Organisation-level quality system and audits
ISO/IEC/IEEE 90003	Guidelines for applying ISO 9001 to software	Interpreting ISO 9001 for software work
ISO/IEC/IEEE 12207	Software life cycle processes	A shared vocabulary for life cycle activities
CMMI (V3.0, 2023)	Practice areas for capability and performance improvement	Improvement roadmaps and judging vendor appraisals
ISTQB syllabi	Tester certification at Foundation, Advanced and Expert levels	A structured learning path for testers
SWEBOK Guide	IEEE Computer Society guide to the software engineering body of knowledge	A map of the whole discipline, including software quality

Source: ISO, IEC, IEEE, CMMI Institute (ISACA), ISTQB and IEEE Computer Society publications

Six principles to take back to work

- **Build quality in:** Reviews and prevention cost less than finding defects late in test or in the field.
- **Evidence over opinion:** Records, metrics and audits turn 'we did it' into something others can verify.
- **Tailor, never skip:** Small, urgent or prototype work still follows standards, just tailored ones.
- **Control change:** Every change to a baseline is requested, approved, tested and recorded.
- **Improve the process:** Treat every escaped defect as a lesson for the process, not a reason to blame a person.
- **Stay sceptical:** Question methods, tools and models, and adopt what the evidence supports in your context.

Where the capstone sits in KRISA 2.0

Because it spans the full lifecycle, the capstone draws on the handbook's final chapters, acceptance, implementation and DevOps alignment.

[Ch.6](#) • Acceptance Testing

Master Test Plan; UAT & PAT, the execution backbone the capstone walks through.

Modules 4, 7

[Ch.7](#) • Implementation

FAT, data migration, user manual, system handover, mirrored in the capstone's audit & documentation steps.

Modules 7-8

[Ch.8](#) • DevOps Alignment

Automation, CI/CD, monitoring; KRISA and [PPrISA](#) and DevOps mapping, the capstone's change & monitoring steps.

Modules 5, 8

Based on KRISA 2.0 Beta 2026 (chapters dated 15 Jan 2026). KRISA 1.0 remains the official baseline until 2.0 is finalized, confirm the version with your agency.

Abbreviations used in this module (1 of 5)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
AI	Artificial Intelligence	Software that can answer questions and write text or code by learning patterns from large amounts of data.
API	Application Programming Interface	A doorway that lets one program ask another program for data or an action, without using a screen.
APM	Application Performance Monitoring	Tools that watch a live system and show how fast and healthy it is.
BRS	Business Requirements Specification	The document that states what the business needs, written in business language. KRISA deliverable D02.
BVA	Boundary Value Analysis	Testing at the edges of the allowed values, such as 0, 1, 10 and 11, because mistakes often hide there.
CAPA	Corrective and Preventive Action	Fixing a problem that has happened, then changing the process so it does not happen again.
CCB	Change Control Board	The group that approves, rejects or defers requests to change an approved baseline.
CI	Continuous Integration, or Configuration Item	Two meanings. In DevOps, every code change is built and tested automatically. In configuration management, it is any controlled item such as a document or code file.
CI/CD	Continuous Integration and Continuous Delivery	An automated pipeline that builds, tests and prepares every change for release.

Abbreviations used in this module (2 of 5)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
CMM	Capability Maturity Model	An older five-level model that rates how mature and repeatable an organisation's software process is.
CMMI	Capability Maturity Model Integration	The newer model that replaced CMM. Used to improve and appraise how an organisation works.
DPP	Dokumen Permulaan Projek	The Project Initiation Document. PPrISA template PPrISA01.
EP	Equivalence Partitioning	Grouping inputs that behave the same way, then testing one value from each group.
FAT	Factory Acceptance Test	The formal test with the customer before the system is installed at the live site.
ID	Identifier	A unique label, such as REQ-CHK-01, so that each item can be traced.
IEC	International Electrotechnical Commission	An international standards body that publishes many software standards together with ISO.
IEEE	Institute of Electrical and Electronics Engineers	A professional body that publishes software engineering standards such as IEEE 730 and IEEE 829.
ISACA	ISACA, formerly the Information Systems Audit and Control Association	The professional body for IT audit and governance. It now owns CMMI.

Abbreviations used in this module (3 of 5)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
ISO	International Organization for Standardization	Publishes international standards such as ISO 9001 and ISO/IEC 25010.
ISTQB	International Software Testing Qualifications Board	Offers the best-known certifications for software testers.
JDN	Jabatan Digital Negara	The Malaysian government department that publishes KRISA and PPrISA.
KRISA	Kejuruteraan Sistem Aplikasi Sektor Awam	The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition.
MIT	Massachusetts Institute of Technology	A university in the United States. Source of the Leveson course notes.
PAT	Provisional Acceptance Test	The KRISA acceptance test done before provisional acceptance of the system, including performance checks.
PDF	Portable Document Format	A document file that looks the same on every computer.
PPP	Pelan Pengurusan Projek	The Project Management Plan. PPrISA template PPrISA02.
PPrISA	Panduan Pengurusan Projek ICT Sektor Awam	The Malaysian public sector guide for managing ICT projects. Codes such as PPrISA05 are its templates.

Abbreviations used in this module (4 of 5)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
PS	PowerShell prompt	When a slide shows PS>, type that command in Windows PowerShell.
QA	Quality Assurance	Preventing defects by setting up good processes and checking that people follow them.
RTM	Requirements Traceability Matrix	A table linking each requirement to its design, tests and defects, so nothing is missed.
SAT	Site Acceptance Test	The formal acceptance test carried out at the live site.
SCM	Software Configuration Management	Tracking every version and change so the right items are built and released.
SDLC	Software Development Life Cycle	The phases a system goes through, from planning to maintenance.
SDS	Spesifikasi Reka Bentuk Sistem	The System Design Specification. KRISA deliverable D04.
SQA	Software Quality Assurance	All the planned work that gives confidence the software meets its quality requirements.

Abbreviations used in this module (5 of 5)

Every short form on these slides, explained in plain words. Keep this page handy during the lab.

Abbreviation	Stands for	What it means in plain words
SQC	Software Quality Control	Checks on the product itself: reviews, tests and audits.
SQM	Software Quality Metrics	Numbers used to measure quality, such as defect density.
SQP	Software Quality Plan	The plan stating which standards, reviews, tests and audits a project will use.
SRS	Spesifikasi Keperluan Sistem	The System (or Software) Requirements Specification: what the system must do. KRISA deliverable D03.
SWEBOK	Software Engineering Body of Knowledge	An IEEE guide that maps the whole software engineering discipline.
TDD	Test-Driven Development	Write a failing test first, then write just enough code to make it pass.
UAT	User Acceptance Test	Users try the system to confirm it meets their needs before they accept it.
UI	User Interface	The screens, buttons and forms that people use.

References and templates for this module

Every line opens the official JDN document at the exact page. Templates download as Word or PDF.

WHERE IT IS IN THE GUIDES

Bab 8, Serahan mengikut fasa (penyelarasan DevOps)
page 8 · KRISAv2 Bab 8 Penyelarasan DevOps

Bab 1, 1.6 Faktor Kejayaan
page 14 · KRISAv2 Bab 1 Perancangan

PPrISA 2.0, Rajah 2.4 Pelan Tindakan (templat mengikut fasa)
page 21 · PPrISA 2.0 Versi Beta Feb 2025

PPrISA 2.0, 6.1.2 Penilaian Akhir
page 98 · PPrISA 2.0 Versi Beta Feb 2025

TEMPLATES TO USE

D03 Spesifikasi Keperluan Sistem (SRS)	WORD	PDF
D12 Pelan Induk Pengujian	WORD	PDF
D13 Pelan Ujian Penerimaan (UAT/PAT)	WORD	PDF
D14 Laporan Ujian Penerimaan (UAT/PAT)	WORD	PDF
D16 Laporan Penamatan Ujian	WORD	PDF
PPrISA05 Pelan Pengurusan Kualiti	WORD	PDF
PPrISA07 Pelan Pengurusan Risiko	WORD	PDF
PPrISA10 Borang Pindaan	WORD	PDF

— COURSE COMPLETE

Thank you for building quality in.

Two days, one project, nine activities, a complete quality trail you built yourselves. Certificates follow the capstone presentation.



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SQA • 2-DAY HANDS-ON WORKSHOP • v5 (KRISA 2.0)

M09 · W09 · 10 MINUTES · Hari 2, 17:10 hingga 17:20

Hands-On SQA-AI Assistant Workshop

Capstone with an AI pair: prove the quality gate

Let the AI evaluate a quality gate from the evidence you produced over two days, including a live test run, then present the trail and the AI mistakes you caught.

TOOL

opencode 1.18 in the lab folder · model: Big Pickle via OpenCode Zen (free, no API key) · synthetic ShopFast data only

[Step-by-step guide \(web\)](#)[Workshop slides \(W09\)](#)[Setup guide W00](#)[opencode docs](#)

Capstone with an AI pair: prove the quality gate

STEPS

- 1 Run /quality-gate G3 with a live test run
- 2 Check every criterion has an evidence file
- 3 Ask the AI to list its own weakest conclusions
- 4 Commit your outputs as audit evidence

OUTPUT TO SAVE

`outputs/m09-quality-gate.md`

TYPE THIS (opencode> in opencode, PS> in PowerShell)

```
opencode> /quality-gate G3 jalankan-ujian
```

```
opencode> Senaraikan 3 kesimpulan paling lemah dalam keputusan quality gate tadi dan bukti tambahan yang diperlukan untuk setiap satu. Jangan tulis fail.
```

```
PS> git add -f outputs shopfast/tests/ui/m05-senario.spec.js ; git commit -m "bukti kemuncak pasukan"
```

HUMAN CHECK BEFORE YOU ACCEPT IT

- Every criterion links to an evidence file and a real value
- No GO while any criterion is unmet or without evidence
- At least one AI mistake was found and corrected by the team

[Open step-by-step guide](#)

[Module hub](#)

Pembantu AI dalam SQA: AI mendraf, manusia bertanggungjawab

AI assistant in SQA: AI drafts, humans remain accountable

Aliran kerja setiap bengkel Hands-On SQA-AI Assistant Workshop menggunakan opencode

1

Konteks

@SRS, @templat,
AGENTS.md

2

Prompt

arahan /review-srs

3

Draf AI

opencode + model

4

Semak manusia

banding dengan templat
KRISA/PPrISA

5

Simpan bukti

outputs/ + git commit

Guna AI untuk

Draf pelan dan laporan, cadang kes ujian
EP/BVA, jana skrip ujian, ringkaskan log, cari
jurang RTM.

Jangan

Tampal data sebenar agensi, terima output
tanpa semak, biar AI reka ID keperluan, kongsi
sesi awam.

Bukti audit

Simpan prompt, output dan semakan dalam git
supaya jejak audit lengkap (Modul 7).